



Q28. State Biot-Savart law, giving the mathematical expression for it.

Use this law to derive the expression for the magnetic field due to a straight conductor carrying current at a point along its axis.

How does a circular loop carrying current behave as a magnet?

Answer: Statement of Biot Savart Law: The magnitude of the magnetic field due to current element is directly proportional to the current, the element length and inversely proportional to the square of the distance of the field point. The direction of magnetic field is perpendicular to the plane containing current element and distance.

	Given i = current flowing through the element dl = the length of the element of the conductor $r = AP$ = the distance of the given point from the element of the conductor $\hat{r}$ = a unit vector along AP $\hat{dl}$ = a vector of unit length along the axis of the element in the direction of current θ = the angle between $\hat{r}$ and $\hat{dl}$ Let $\vec{dB}$ = magnetic induction vector at the point P due to the current element Experimentally it has been found that $ \vec{dB} \propto i, \vec{dB} \propto dl, \vec{dB} \propto \frac{1}{r^2}, \vec{dB} \propto \sin \theta$ $dB \propto \frac{idl \sin \theta}{r^2}$ $\vec{dB} = \frac{\mu_0}{4\pi} \frac{i}{r^2} (\vec{dl} \times \hat{r})$ $\vec{dB} = \frac{\mu_0}{4\pi} \frac{i}{r^3} (\vec{dl} \times \vec{r})$
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To find the magnetic field at a point due to an infinitely long straight conductor carrying current.

	Let $AP = r, \angle OAP = \theta$ Applying Biots law the magnetic induction vector at P due to the current element at A $\vec{dB} = \frac{\mu_0}{4\pi} \frac{i}{r^2} \vec{dx} \times \hat{r} \rightarrow (1)$ The magnitude of this induction vector $dB = \frac{\mu_0}{4\pi} \frac{i}{r^2} dx \sin \theta \rightarrow (2)$
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$$B = \int dB = \int \frac{\mu_0}{4\pi} \frac{i}{r^2} dx \sin \theta \rightarrow (3)$$

Equation (3) contains three variables x , r & θ hence it is represented in terms of one variable only.

	$\operatorname{cosec} \theta = \frac{r}{a}, r = a \operatorname{cosec} \theta$ When $x = -\infty, \theta = 0$ and when $x = +\infty, \theta = 180^\circ = \pi$ $\therefore B = \frac{\mu_0}{4\pi} i \int_0^\pi \frac{a \operatorname{cosec}^2 \theta \sin \theta}{a^2 \operatorname{cosec}^2 \theta} d\theta$ $B = \frac{\mu_0}{4\pi} \frac{i}{a} \int_0^\pi \sin \theta d\theta = \frac{\mu_0}{4\pi} \frac{i}{a} [-\cos \theta]_0^\pi = \frac{\mu_0}{4\pi} \frac{2i}{a} \rightarrow (5)$
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Equation(5) is known as Biot-Savart formula and gives the magnitude of the magnetic field at a point near a straight conductor carrying current. The direction of the magnetic field which was found to be along Z axis in the figure can be stated as follows:

“Grasp the conductor in right hand with the thumb in the direction of current. The curl of the finger tips gives the direction of the magnetic field.”

	The direction of magnetic field is along the axis of the coil which can be found as follows “Curl the finger of right hand in the direction of current in the coil the thumb gives the direction of the magnetic field”. Thus magnetic flux emitting for this magnetic field produced is along X axis then this side i.e. coil facing +X axis will behave as North pole and negative X axis (X' axis) will behave as South pole. Hence coil (loop) behaves as a magnetic dipole.
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Or,

With the help of labelled diagram, state the underlying principle of a cyclotron. Explain clearly how it works to accelerate the charged particles.

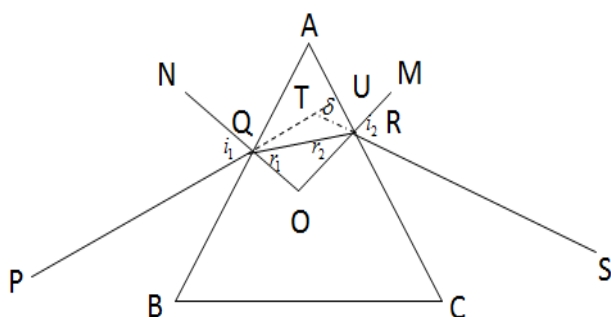
Show that cyclotron frequency is independent of energy of the particle. Is there an upper limit on the energy acquired by the particle? Give reason.

Q29. (a) Draw a ray diagram to show refraction of a ray of monochromatic light passing through a glass prism.

Deduce the expression for the refractive index of a glass in terms of angle of prism and angle of minimum deviation.

(b) Explain briefly how the phenomenon of total internal reflection is used in fibre optics.

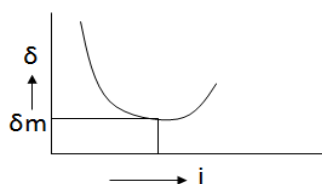
Answer (a): Refraction through prism



A = Angle of prism

Had there been no prism, the ray PQ would have gone straight along PQU, but due to the prism the ray finally goes along RS. Hence the prism has deviated the ray through the angle $UTR = \delta$ and is known as angle of deviation.

ON & OM are perpendiculars drawn on AB and AC respectively.



Let

$\angle PQN = i_1$ = angle of incidence

$\angle RQO = r_1$

$\angle SRM = i_2$ = angle of emergence

$\angle AQO = \pi/2 = \angle ARO$

In quadrilateral AQOR

Sum of two opposite angles $\angle AQO + \angle ARO = 180^\circ$

Hence AQOR is a co cyclic quadrilateral

$\angle QAR + \angle QOR = 180^\circ$

$A + \angle QOR = 180^\circ \rightarrow (1)$

In ΔQOR

$r_1 + r_2 + \angle QOR = 180^\circ \rightarrow (2)$

From equation (1) and equation (2):

$A = r_1 + r_2 \rightarrow (3)$

$\angle TQR = i_1 - r_1$

$\angle TRQ = i_2 - r_2$

$\delta = (i_1 - r_1) + (i_2 - r_2)$

$\delta = (i_1 + i_2) - (r_1 + r_2)$

$\delta = (i_1 + i_2) - A \rightarrow (4)$



From equation (4) we find that the angle of deviation δ depends on the angle of Incidence. Experimentally it is found that as the angle of incidence increases angle of deviation decreases, reaches a minimum value δ_m for a particular angle of incidence and then increases again.

The minimum value δ_m called the angle of minimum deviation. It can be analytically proved that when

$i_1 = i_2, r_1 = r_2$ i.e. the path of the ray through the prism is parallel to the base of the prism then only the angle of deviation is minimum.

Condition for minimum deviation

$$i_1 = i_2 \text{ \& \; } r_1 = r_2$$

From equation (4)

$$\delta_m = i + i - A$$

$$i = \frac{A + \delta_m}{2}$$

$$A = r_1 + r_2 = r + r$$

$$A = 2r$$

$$r = \frac{A}{2}$$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin \frac{A}{2}}$$

For a thin prism the angle of prism A is very small

$$\sin \frac{A}{2} \approx \frac{A}{2}$$

$$\sin\left(\frac{A + \delta_m}{2}\right) \approx \frac{A + \delta_m}{2} \approx \frac{A + \delta}{2}$$

$$\therefore \mu = \frac{\frac{A + \delta}{2}}{\frac{A}{2}}$$

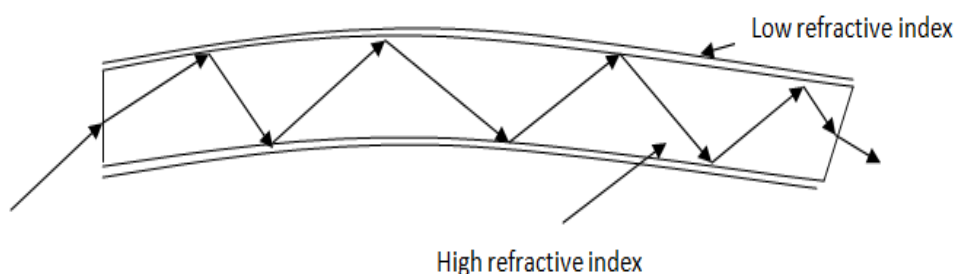
$$\text{or } A\mu = A + \delta$$

$$\text{or } \delta = A(\mu - 1) \rightarrow (6)$$

Equation (6) gives the deviation produced by thin prism.



- (b) Total internal reflection is used in optical fibre: Optical fibre consists of a core and cladding. Refractive index of the material of the core is higher than that of cladding. When a signal, in the form of light, is directed into the optical fibre, at an angle greater than the respective critical angle, it undergoes repeated total internal reflections as shown along the length of the fibre.



Finally the signal comes out from the other end with almost negligible loss of intensity.